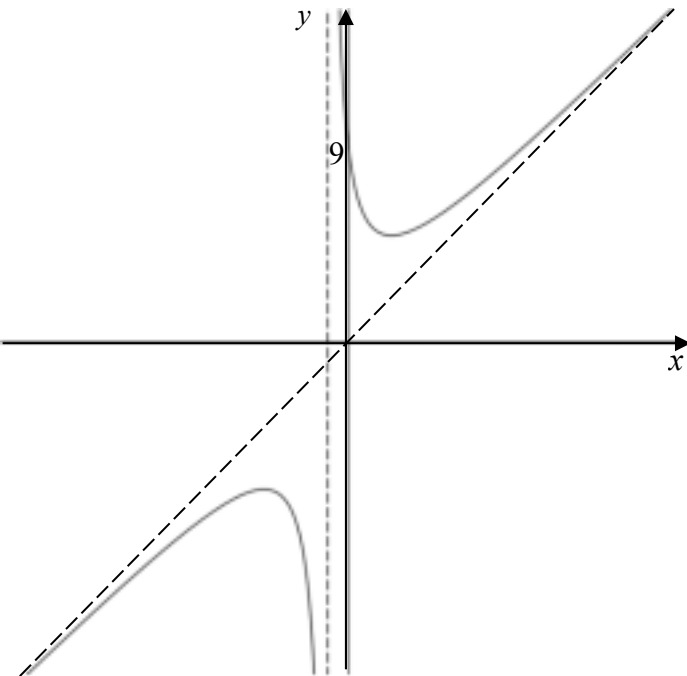
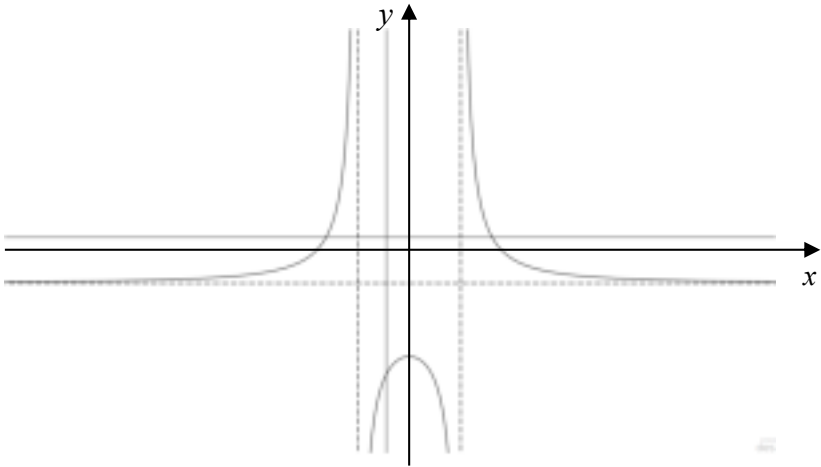


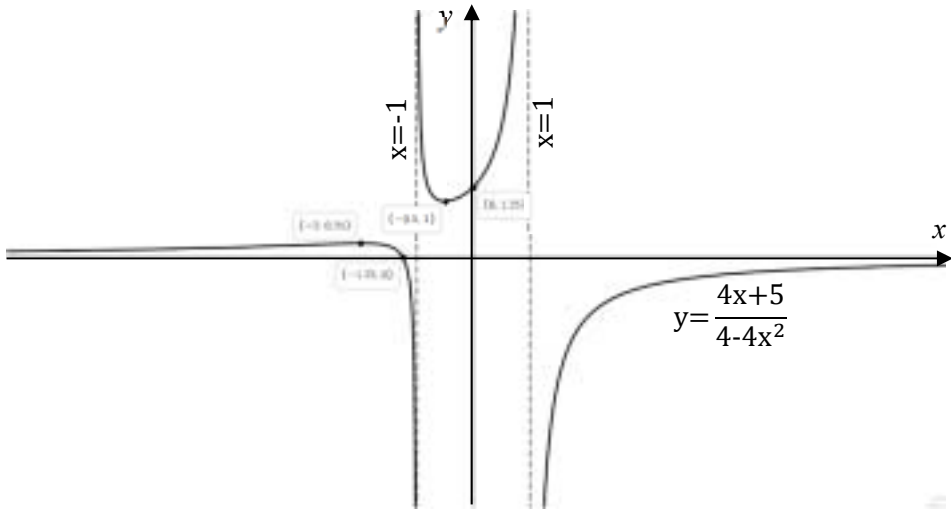
Question	Answer	Marks	Guidance
7(a)	$x = -1$	B1	States vertical asymptote.
	$y = \frac{x(x+1)+9}{x+1} = x + \frac{9}{x+1}$	M1	Finds oblique asymptote.
	$y = x$	A1	
		3	
7(b)	$\frac{dy}{dx} = 1 - 9(x+1)^{-2} = 0 \Rightarrow (x+1)^2 = 9$	M1 A1	Differentiates and sets derivative equal to 0.
	$(2, 5)$	A1	
	$(-4, -7)$	A1	
		4	

Question	Answer	Marks	Guidance
7(c)		B1	Axes labelled and correct asymptotes drawn.
		B1	Upper branch with $(0, 9)$ stated or shown on diagram.
		B1	Lower branch correct and good approach to asymptotes throughout, no extra branches.
		3	

Question	Answer	Marks	Guidance
7(d)		B1 FT	FT from sketch in (c) with asymptotes shown.
	$x^2 + x + 9 = \frac{13}{2}(x+1) \text{ or } x^2 + x + 9 = -\frac{13}{2}(x+1)$ $x^2 - \frac{11}{2}x + \frac{5}{2} = 0 \text{ or } x^2 + \frac{15}{2}x + \frac{31}{2} = 0$	M1 M1	Finds critical points, award M1 for each case. May state that $x^2 + x + 9 = -\frac{13}{2}(x+1)$ has no real solutions since $7 > \frac{13}{2}$.
	$x = \frac{1}{2}, 5$	A1	
	$x < \frac{1}{2}$ and $x > 5$.	A1	
		5	

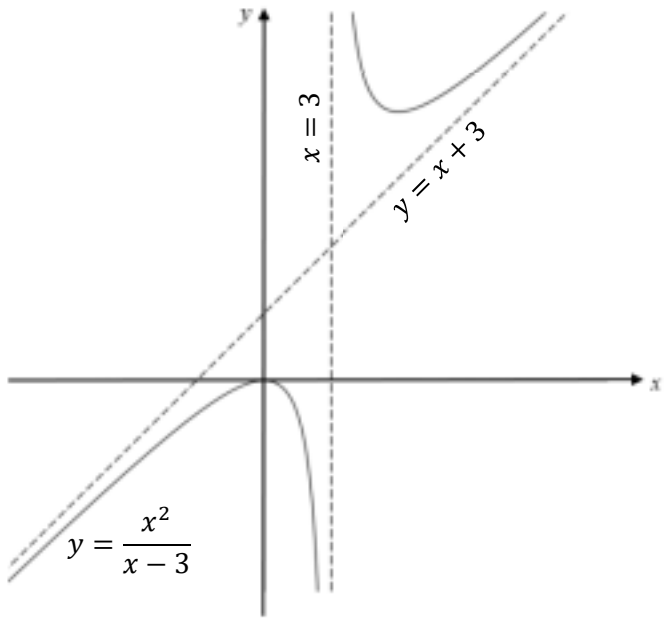
Question	Answer	Marks	Guidance
7(a)	$x = \frac{1-\sqrt{5}}{2}, x = \frac{1+\sqrt{5}}{2}$	B1	Vertical asymptotes. Must be exact.
	$y = -1$	B1	Horizontal asymptote.
		2	
7(b)	$\frac{dy}{dx} = \frac{(1+x-x^2)(2x-1) - (x^2-x-3)(1-2x)}{(1+x-x^2)^2}$	M1	Finds $\frac{dy}{dx}$.
	$(2x-1)(-2) = 0$	M1	Sets their $\frac{dy}{dx}$ equal to 0 and forms equation.
	$(\frac{1}{2}, -\frac{13}{5})$	A1	WWW
		3	
7(c)		B1	Both axes labelled and correct asymptotes shown.
		B1	Correct shape and position, with all asymptotic behaviour clear.
	$(\frac{1}{2} + \frac{1}{2}\sqrt{13}, 0), (\frac{1}{2} - \frac{1}{2}\sqrt{13}, 0), (0, -3)$	B1	States exact coordinates of intersections with axes.
		3	

Question	Answer	Marks	Guidance
7(d)		B1 FT	FT from sketch in (c).
		B1	Correct shape as x tends to infinity and intersections with x axis.
	$\frac{x^2 - x - 3}{1 + x - x^2} = 3 \quad \text{or} \quad \frac{x^2 - x - 3}{1 + x - x^2} = -3$ $4x^2 - 4x - 6 = 0 \quad \text{or} \quad -2x^2 + 2x = 0$	M2	Finds critical points, award M1 for each case.
	$x = \frac{1}{2} + \frac{1}{2}\sqrt{7}, \quad x = \frac{1}{2} - \frac{1}{2}\sqrt{7}, \quad x = 0, \quad x = 1$	A1	Must be exact.
	$x < \frac{1}{2} - \frac{1}{2}\sqrt{7}, \quad 0 < x < 1, \quad x > \frac{1}{2} + \frac{1}{2}\sqrt{7}$	A1 FT	Must be three distinct regions and strict inequalities.
		6	

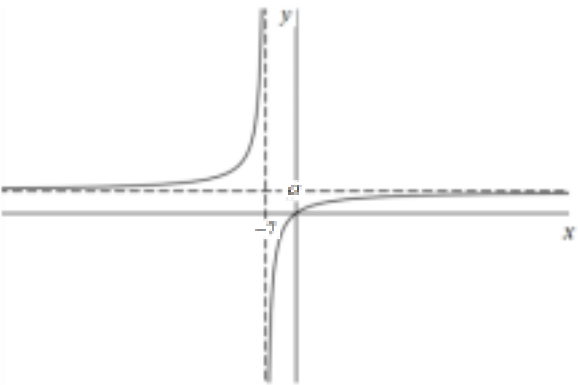
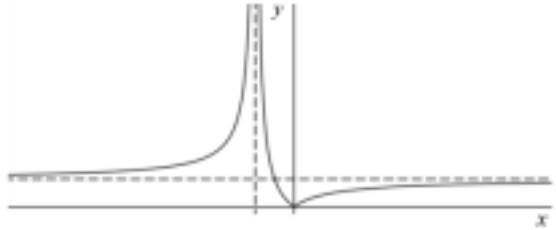
Question	Answer	Marks	Guidance
7(a)	$x = 1, x = -1$	B1	Vertical asymptotes.
	$y = 0$	B1	Horizontal asymptote.
		2	
7(b)	$\frac{dy}{dx} = \frac{(4-4x^2)(4) - (4x+5)(-8x)}{(4-4x^2)^2}$	M1	Finds $\frac{dy}{dx}$.
	$16x^2 + 40x + 16 = 0$	M1	Sets equal to 0 and forms equation.
	$(-2, \frac{1}{4}), (-\frac{1}{2}, 1)$	A1 A1	
		4	
7(c)		B1	Axes and asymptotes.
		B1	Correct shape and position.
	$(-\frac{5}{4}, 0), (0, \frac{5}{4})$	B1	States coordinates of intersections with axes.
		3	

Question	Answer	Marks	Guidance
7(d)		B1 FT	FT from sketch in part (c).
		B1	Correct shape at infinity.
	$\frac{4x+5}{4-4x^2} = \frac{5}{4} \text{ or } \frac{4x+5}{4-4x^2} = -\frac{5}{4}$ $5x^2 + 4x = 0 \text{ or } 5x^2 - 4x - 10 = 0$	M2	Finds critical points, award M1 for each case.
	$x = -\frac{4}{5}, \quad x = 0 \quad \text{or} \quad x = \frac{2}{5} - \frac{3}{5}\sqrt{6}, \quad x = \frac{2}{5} + \frac{3}{5}\sqrt{6}$	A1	A0 if $-1.07, 1.87$
	$\frac{2}{5} - \frac{3}{5}\sqrt{6} < x < -\frac{4}{5}, \quad 0 < x < \frac{2}{5} + \frac{3}{5}\sqrt{6}, \quad x \neq \pm 1$	A1 FT	Condone exclusion of $x = \pm 1$ from the range.
		6	

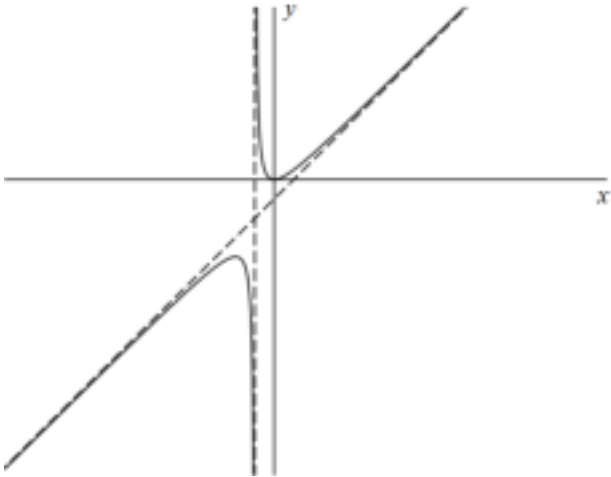
Question	Answer	Marks	Guidance
6(a)	$x = 3$	B1	States vertical asymptote.
	$y = x + 3 + \frac{9}{x-3}$ leading to $y = x + 3$	M1 A1	Finds oblique asymptote.
		3	
6(b)	$yx - 3y = x^2$ leading to $x^2 - yx + 3y = 0$	M1 A1	Forms quadratic in x .
	$y^2 - 4(3y) < 0$ leading to $y^2 - 12y < 0$	M1	Uses that discriminant is negative.
	$0 < y < 12$	A1	AG
		4	

Question	Answer	Marks	Guidance
6(c)		B1	Axes and asymptotes.
		B1	Branches correct.
		2	

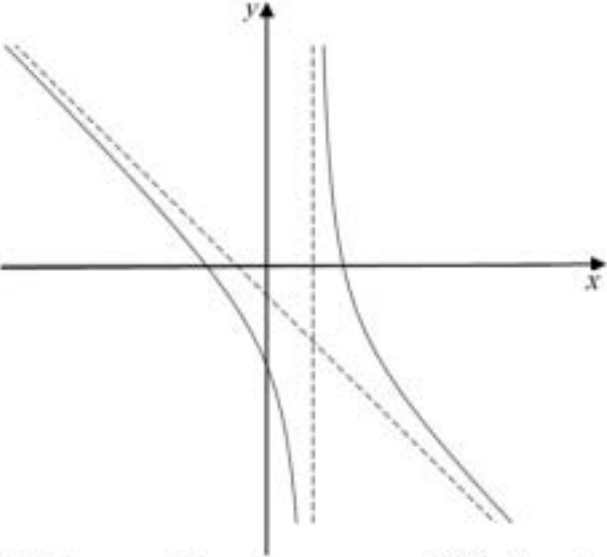
Question	Answer	Marks	Guidance
6(d)(i)	Graph showing the function $y = \left \frac{x^2}{x-3} \right$ and the function $y = x - 3$. The graph includes the vertical asymptote $x = 3$ and the x-intercepts $(-3, 0)$ and $(3, 0)$. The y-intercept of $y = x - 3$ is $(0, -3)$.	B1 FT	FT from sketch in (c).
		B1	Correct shape at infinity.
		B1	Correct shape of $y = x - 3$.
		B1	Correct intercepts with axes (may be seen on graph).
		4	
6(d)(ii)	$c \leq -3$	B1	
		1	

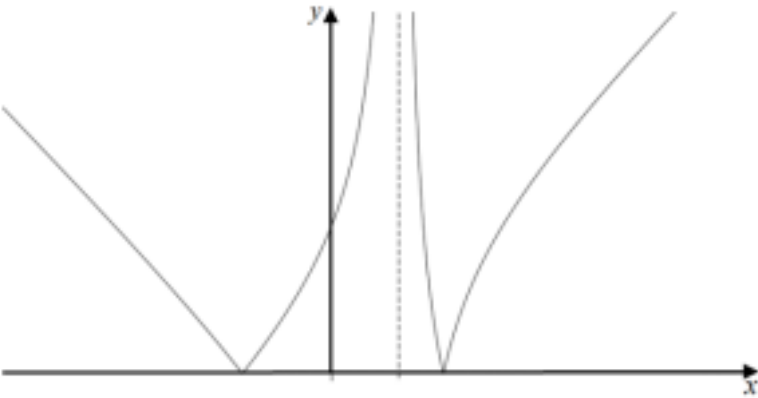
Question	Answer	Marks
1(a)	 (B1 for axes and asymptotes correct, B1 for branches correct)	B1
		B1
		2
1(b)	 (B1 FT from sketch in part (a))	B1 FT
	$\frac{ax}{x+7} = \frac{a}{2}$ or $\frac{ax}{x+7} = -\frac{a}{2}$	M1
	$x = 7$ and $x = -\frac{7}{3}$	A1
	$x < -7$, $-7 < x < -\frac{7}{3}$, $x > 7$	A1
		4

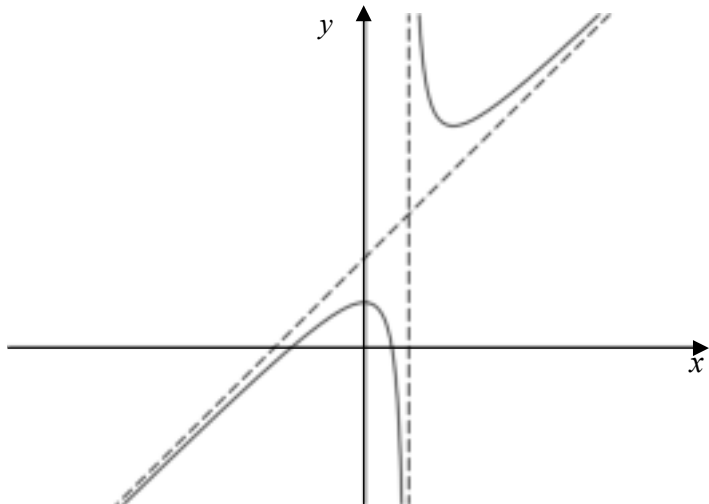
Question	Answer	Marks
3(a)	$x = -\frac{1}{2}$	B1
	$y = \frac{(2x+1)(\frac{1}{2}x - \frac{1}{4}) + \frac{1}{4}}{2x+1} = \frac{1}{2}x - \frac{1}{4} + \frac{1}{4(2x+1)}$	M1
	$y = \frac{1}{2}x - \frac{1}{4}$	A1
		3
3(b)	$\frac{dy}{dx} = \frac{1}{2} - \frac{1}{2(2x+1)^2} = 0 \Rightarrow (2x+1)^2 = 1$	M1
	$x = 0, -1$	A1
	$(0, 0), (-1, -1)$	A1
		3

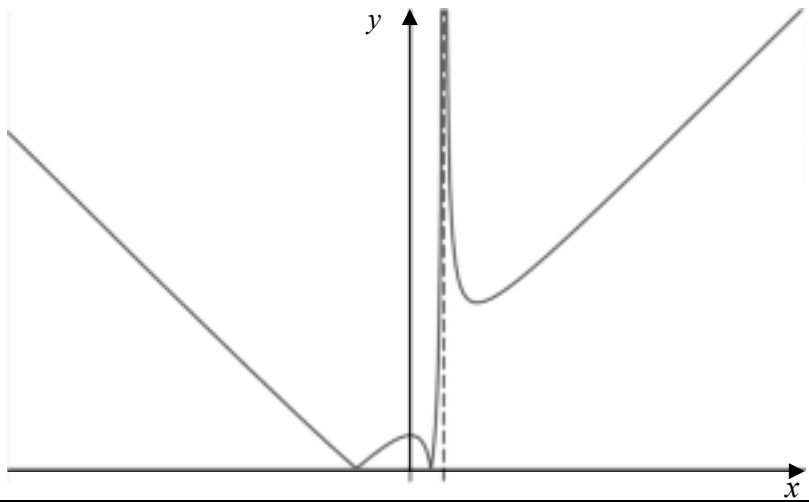
Question	Answer	Marks
3(c)	 (B1 for axes and asymptotes correct, B1 for upper branch correct, B1 for lower branch correct)	B1
		B1
		B1
		3

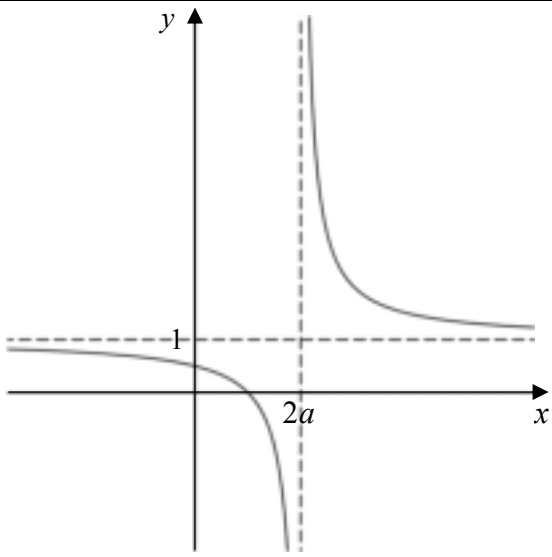
Question	Answer	Marks
6(a)	$x = \frac{3}{2}$	B1
	$-2x^2 + x + 10 = (2x - 3)(-x - 1) + 7 \Rightarrow y = -x - 1$	M1 A1
		3
6(b)	$\frac{dy}{dx} = \frac{(2x-3)(1-4x) - 2(10+x-2x^2)}{(2x-3)^2}$	M1
	$4x^2 - 12x + 23 = 0 \quad \left(\text{or } \frac{dy}{dx} = -1 - \frac{14}{(2x-3)^2} \right)$	A1
	$12^2 - 4(4)(23) = -224 < 0 \text{ (or } y' < 0 \text{)} \Rightarrow \text{No turning points.}$	A1
		3

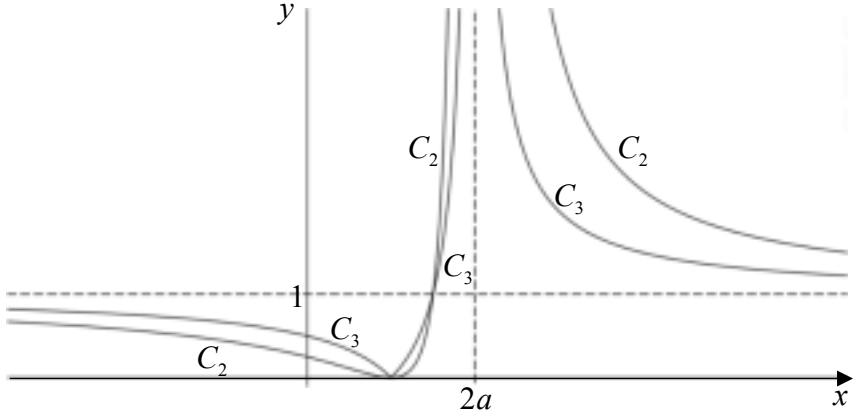
Question	Answer	Marks
6(c)		B1
		B1
	(B1 for axes and asymptotes correct, B1 for branches correct)	
	$(0, -\frac{10}{3}), (-2, 0), (\frac{5}{2}, 0)$	B1
		3

Question	Answer	Marks
6(d)	 (B1 FT for their sketch in (c), B1 for correct shape at infinity)	B1 FT
		B1
	$\frac{10+x-2x^2}{2x-3} = 4 \text{ or } \frac{10+x-2x^2}{2x-3} = -4$ $2x^2 + 7x - 22 = 0 \text{ or } 2x^2 - 9x + 2 = 0$	M2
	$x = -\frac{11}{2}, 2 \text{ or } x = \frac{1}{4}(9 \pm \sqrt{65})$	A1
	$-\frac{11}{2} < x < \frac{1}{4}(9 - \sqrt{65}) \text{ and } 2 < x < \frac{1}{4}(9 + \sqrt{65})$	A1 FT
		6

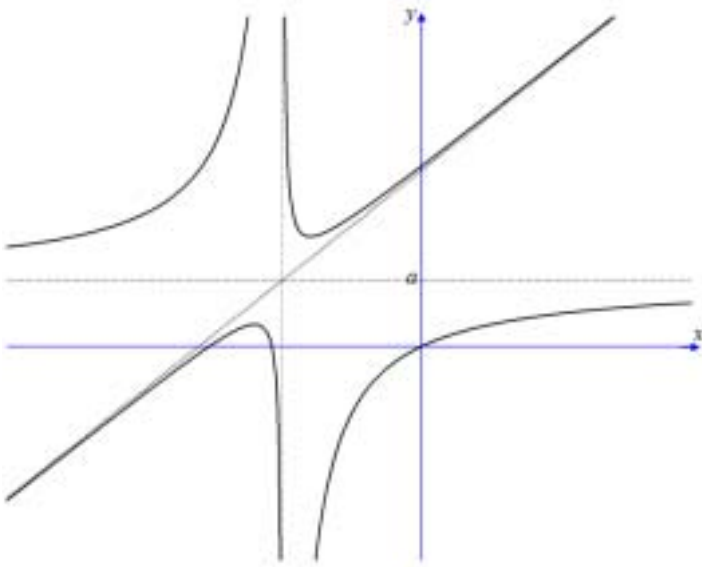
Question	Answer	Marks	Guidance
6(a)	$x = 1$	B1	States vertical asymptote.
	$x^2 + x - 1 = (x - 1)(x + 2) + 1 \Rightarrow y = x + 2$	M1 A1	Finds oblique asymptote.
		3	
6(b)	$yx - y = x^2 + x - 1 \Rightarrow x^2 + (1 - y)x + y - 1 = 0$	M1 A1	Forms quadratic in x .
	$(1 - y)^2 - 4(y - 1) < 0 \Rightarrow y^2 - 6y + 5 < 0$	M1	Uses that discriminant is negative if there are no values of x .
	$1 < y < 5$	A1	AG
		4	
6(c)		B1	Axes and asymptotes.
		B1	Branches correct.
	$(0, 1), \left(-\frac{1}{2} + \frac{1}{2}\sqrt{5}, 0\right), \left(-\frac{1}{2} - \frac{1}{2}\sqrt{5}, 0\right)$	B1	States coordinates of intersections with axes, can be labelled on graph. Accept $(0.618, 0)$ and $(-1.62, 0)$.
		3	

Question	Answer	Marks	Guidance
6(d)		B1 FT	FT from sketch in (c) with both branches.
		B1	Correct shape at extremities.
		2	

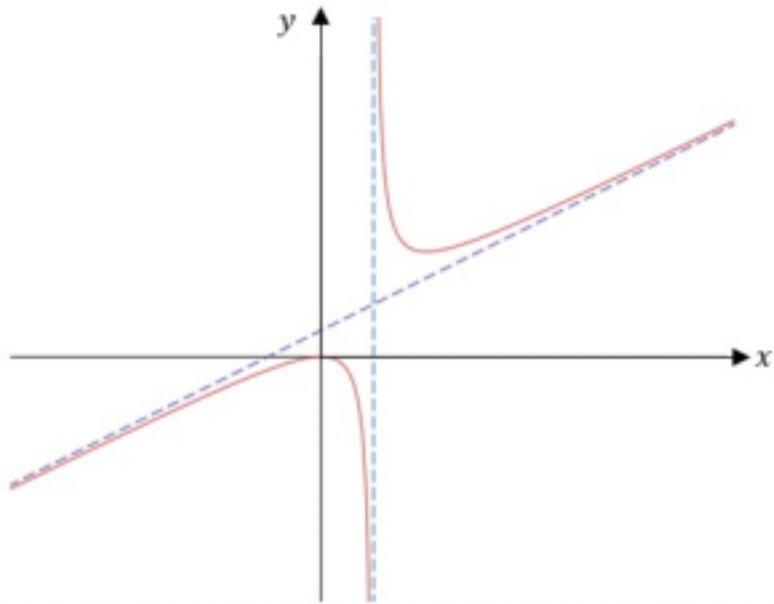
Question	Answer	Marks	Guidance
6(a)		B1	Asymptotes labelled.
		B1	Correct position and shape. Not too truncated.
		2	
6(b)(i)	$\frac{dy}{dx} = -\frac{2a(x-a)}{(x-2a)^3} = 0$	M1	Sets $\frac{dy}{dx} = 0$.
	Solves for x .	M1	
	Since $x \neq 2a$, stationary point is $(a, 0)$.	A1	
		3	

Question	Answer	Marks	Guidance
6(b)(ii)	$(x-a)^2 = (x-a)(x-2a)$ or $(x-a)^2 = -(x-a)(x-2a)$	M1	Finds a critical point.
	$x = a$ and $\frac{3}{2}a$ or $(a, 0)$ or $(\frac{3}{2}a, 1)$	A1	Both x values or one correct point.
	$(a, 0)$ and $(\frac{3}{2}a, 1)$	A1	
		3	
6(c)		B1	Axes and asymptotes.
		B1 FT	C_3 FT from their sketch in part (a). Clearly identified.
		B1	Correct shape of C_2 . Clearly identified.
		B1	Relative positions and intersections correct.
	$x \leq \frac{3}{2}a$	B1	Accept algebraic method.
		5	

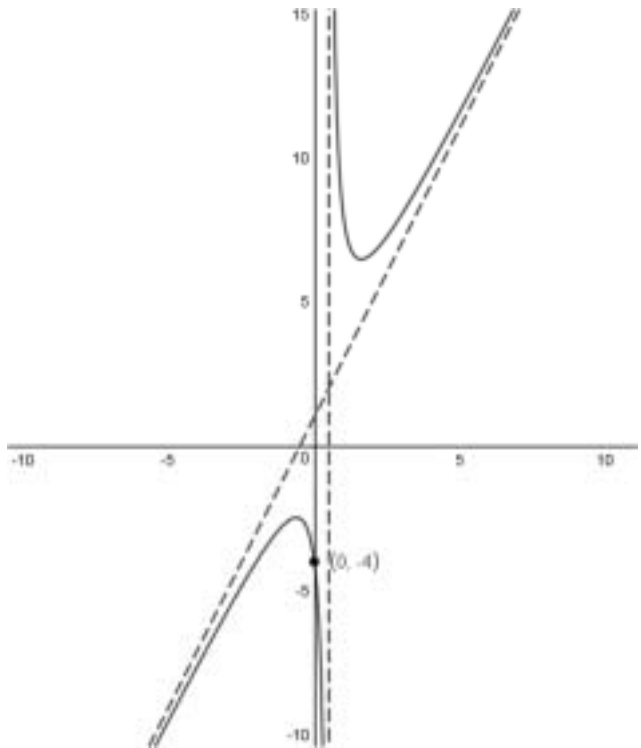
Question	Answer	Marks	Guidance
10(i)	$x = -5$ and $y = a$	B1 B1	
		2	
10(ii)	$x^2 + (a+10)x + 5a + 26 = (x+5)(x+a+5) + 1$	M1	By inspection or long division.
	oblique asymptote is $y = x + a + 5$	A1	
		2	
10(iii)	$x^2 + 10x + 5a + 26 = 0$	M1	Puts y -values equal and forms quadratic equation.
	$10^2 - 4(5a + 26) = -4 - 20a < 0$ so no intersection point	A1	Correct discriminant and conclusion.
		2	
10(iv)	$(x+5)(2x+a+10) - x^2 - ax - 10x - 5a - 26 = 0$	M1	Differentiates and forms quadratic equation.
	$x^2 + 10x + 24 = 0$	A1	
	Stationary points are $(-4, a+2)$ and $(-6, a-2)$	A1	Must have both points.
		3	

Question	Answer	Marks	Guidance
10(v)		B1	Asymptotes drawn, intersection correct.
		B1	C_1 correct.
		B1	C_2 correct.
		3	

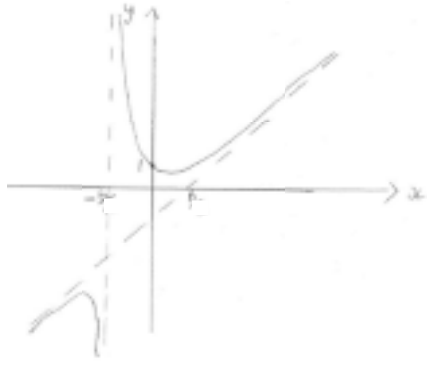
Question	Answer	Marks	Guidance
6(i)	$x = \frac{1}{k}$	B1	States vertical asymptote.
	$y = \frac{(kx-1)(k^{-1}x+k^{-2})+k^{-2}}{kx-1}$	M1	Finds oblique asymptote.
	Oblique asymptote is $y = k^{-1}x + k^{-2}$	A1	
		3	
6(ii)	$y' = \frac{2x(kx-1)-kx^2}{(kx-1)^2} = 0 \Rightarrow kx^2 - 2x = 0$	M1	Differentiates and equates to 0.
	$x = 0, 2k^{-1}$	A1	Finds x -coordinates.
	$(0,0), (2k^{-1}, 4k^{-2})$	A1	Finds y -coordinates
		3	

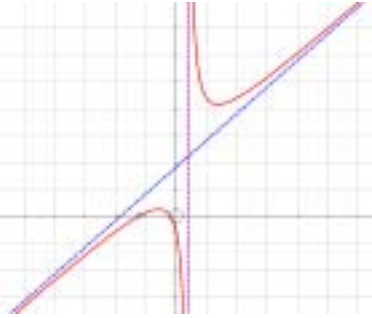
Question	Answer	Marks	Guidance
6(iii)		B1	Axes and asymptotes correct.
		B1	Upper branch correct.
		B1	Lower branch correct. Deduct at most 1 mark for poor forms at infinity.
		3	

Question	Answer	Marks	Guidance
4(i)	$x^2 + 1 = (ax + b)(2x + 1) + c$	M1	Uses that $2x + 1$ is the quotient.
	$\Rightarrow a = \frac{1}{2}, b = -\frac{1}{4}$	A1 A1	
		3	
4(ii)	$x = \frac{1}{2}$	B1 FT	
		1	

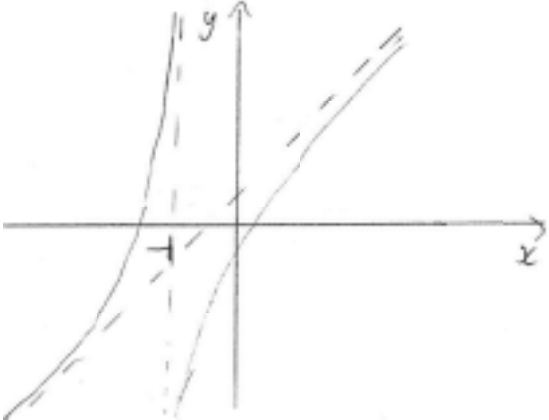
Question	Answer	Marks	Guidance
4(iii)		B1	Intersection (0,-4) given and asymptotes drawn.
		B1	Left branch correct.
		B1 FT	Right branch correct. Deduct at most one mark for poor forms at infinity.
		3	

Question	Answer	Marks	Guidance
6(i)	Vertical asymptote is $x = -b$.	B1	
	$x^2 + b = (x+b)(x-b) + b^2 + b \text{ or}$ $x+b \overline{)x^2 + 0x + b} \quad \begin{array}{r} x-b \\ \hline \end{array}$	M1	By inspection or long division.
	Thus the oblique asymptote is $y = x - b$	A1	
		3	
6(ii)	If $y = 0$ then $x^2 + b = 0$ which has no real root.	B1	Must refer to $b > 0$ OE
		1	
6(iii)	$\frac{dy}{dx} = \frac{2x(x+b) - (x^2 + b)}{(x+b)^2} = 0 \Rightarrow x^2 + 2bx - b = 0$ Or differentiating $y = x - b + \frac{b^2 + b}{x+b}$ and setting $\frac{dy}{dx} = 0$ gives $1 - \frac{b^2 + b}{(x+b)^2} = 0.$	M1	Find $\frac{dy}{dx}$ and set = 0
	$b^2 + b > 0$ Therefore there are two stationary points on C	A1	Use discriminant or $(x+b)^2$ to show two stationary points
		2	

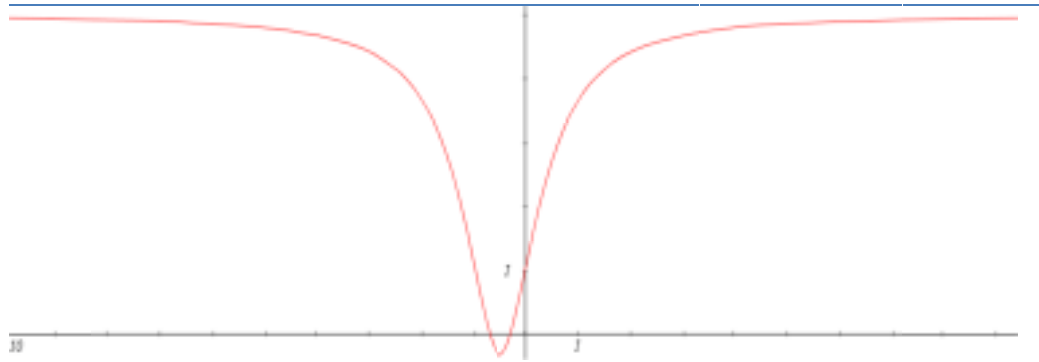
Question	Answer	Marks	Guidance
6(iv)		B1	Intersection (0,1) given and asymptotes drawn
		B1 B1	Each branch correct Penalise at most one mark for poor forms at infinity
		3	

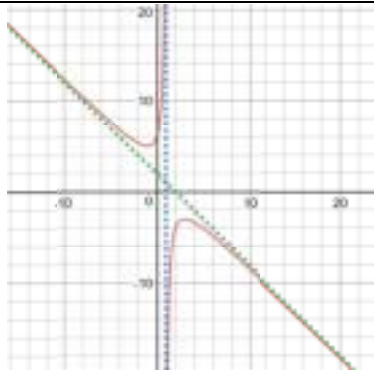
Question	Answer	Marks	Guidance
4(i)	$(-6,0), (-1,0)$	B1	States points of intersection with x -axis.
	$(0,-3)$	B1	States y -intercept
4(ii)	One asymptote is $x = 2$.	B1	
	$y = x + 9 + \frac{24}{x-2} \Rightarrow$ other asymptote is $y = x + 9$.	M1 A1	By inspection or long division. A0 if error in division
4(iii)		B1	Sketches axes and asymptotes, labelled or to scale
		B1	Upper branch correctly located and orientated.
		B1	Lower branch correctly located and orientated. Penalise at most 1 mark for poor forms at infinity
		8	

Question	Answer	Marks	Guidance
6(i)	Vertical asymptote is $x = -1$.	B1	
	$x^2 + ax - 1 = (x+1)(x+a-1) - a$ or $x+1 \overline{)x^2 + ax - 1}$	M1	By inspection or long division.
	Thus the oblique asymptote is $y = x + a - 1$	A1	
		3	
6(ii)	$a^2 + 4 > 0$	B1	
		1	
6(iii)	$\frac{(x+1)(2x+a) - (x^2 + ax - 1)}{(x+1)^2} = 0 \Rightarrow x^2 + 2x + a + 1 = 0$ Discriminant $= 4 - 4(a+1) < 0$	M1	
	Therefore there are no stationary points on C .	A1	
		2	

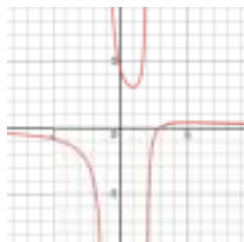
Question	Answer	Marks	Guidance
6(iv)		B1	Correct y-intercept and asymptotes drawn.
		B1 B1	Each branch correct
		3	

Question	Answer	Marks	Guidance
9(i)	$y = 5 - \frac{4}{x^2 + x + 1}$	M1	Alt method: Finding limit
	As $x \rightarrow \pm\infty$, $y \rightarrow 5 \therefore y = 5$ CAO	A1	
		2	
9(ii)	$yx^2 + yx + y = 5x^2 + 5x + 1$ $\Rightarrow (y-5)x^2 + (y-5)x + (y-1) = 0$	B1	Forms quadratic equation in x .
	For real x , $(y-5)^2 - 4(y-5)(y-1) \geq 0$ (condone >)	M1	Uses discriminant
	$\Rightarrow (y-5)(3y+1) \leq 0$	M1	Factorising
	$\Rightarrow -\frac{1}{3} \leq y < 5$, because $y = 5$ is an asymptote (www)	A1	Explaining strict upper inequality (AG)
		4	
9(iii)	$y' = 0 \Rightarrow (x^2 + x + 1)(10x + 5) - (5x^2 + 5x + 1)(2x + 1) = 0$	M1	Differentiates and equates to 0.
	$\Rightarrow 4(2x + 1) = 0 \Rightarrow x = -\frac{1}{2}, y = -\frac{1}{3}$	A1	
		2	

Question	Answer	Marks	Guidance
9(iv)		B1FT	Positive y-intercept at (0,1), FT dep on minimum point from (iii).
		B1	Correct asymptote and completely correct graph.
		2	

Question	Answer	Marks	Guidance
9(i)	$x = 1$	B1	
	$y = 2 - x$	M1A1	
	Total:	3	
9(ii)	$y' = -1 + 4(1 - x)^{-2} = 0$	M1	
	$\Rightarrow x = -1, 3$	A1	
	Turning points are $(-1, 5)$ and $(3, -3)$	A1	
	Total:	3	
9(iii)	$(0, 6)$	B1	
	$y = 0 \Rightarrow x^2 - 3x + 6$; $\Delta = 9 - 24 \Rightarrow$ No intersection with x -axis	B1	
	Total:	2	
9(iv)		B1 FT	Asymptotes correct.
		B1B1	Each branch.
	Total:	3	

Question	Answer	Marks	Guidance
9(i)	Degree of numerator < degree of denominator $\Rightarrow y = 0$ is horizontal asymptote.	B1	
	$(x+1)(x-2)=0 \Rightarrow x=-1$ and $\Rightarrow x=2$ are vertical asymptotes.	B1	
		2	
9(ii)	$yx^2 - (y+3)x + 9 - 2y = 0$	M1	
	No points on C if $(y+3)^2 - 4y(9-2y) < 0$	M1	
	$\Rightarrow 9y^2 - 30y + 9 < 0 \Rightarrow 3y^2 - 10y + 3 < 0$	A1	
	$\Rightarrow (3y-1)(y-3) < 0 \Rightarrow \frac{1}{3} < y < 3$	A1	AG
		4	
9(iii)	$\frac{dy}{dx} = 0 \Rightarrow 3(x^2 - x - 2) - (3x - 9)(2x - 1) = 0$	B1	
	$\Rightarrow \dots \Rightarrow (x-1)(x-5) = 0$	B1	
	$\Rightarrow$ Turning points are $(1,3)$ and $\left(5, \frac{1}{3}\right)$.	B1	
		3	

Question	Answer		Marks	Guidance
9(iv)		Axes, asymptotes and points on axes (0, 4.5) (3,0).	B1	
		RH branch ; Other two branches	B1B1	
			3	

Qu	Solution	Part Marks
7	Vertical asymptote is $x = 2$. $y = x + 2 + \frac{4}{x-2} \Rightarrow$ Oblique asymptote is $y = x + 2$. $y = \frac{x^2}{x-2} \Rightarrow x^2 - yx + 2y = 0$ Quadratic has no real roots (i.e. no points on C) if $\Delta < 0 \Rightarrow y^2 - 8y < 0$ $\Rightarrow y(y-8) < 0 \Rightarrow 0 < y < 8$. (AG) Correct inequality Axes and asymptotes. Each branch, showing (0,0) and (4,8). (Deduct at most 1 mark for poor forms at infinity and/or missing coordinates.)	B1 M1A1 [3] B1 M1 M1A1 [4] B1 ↗ B1B1 [3]

Qu	Solution	Marks
5	$\frac{dy}{dx} = \frac{(x^2 - 9) - 2x(x + 2)}{(x^2 - 9)^2}$ $= \frac{-x^2 - 4x - 9}{(x^2 - 9)^2} = \frac{-(x + 2)^2 - 5}{(x^2 - 9)^2} \Rightarrow \frac{dy}{dx} < 0$ Asymptotes: $x = \pm 3$; $y = 0$ Sketch: Axes and asymptotes ; Outside branches Middle branch, showing $\left(0, -\frac{2}{9}\right)$ and $(-2, 0)$. (Deduct at most 1 mark for poor forms at infinity.)	B1 M1A1 [3] B1B1 [2] B1B1 B1 [3]

Qn & Part	Solution	Marks
10	$-\frac{1}{2} + \frac{(3x-1)^2}{2(x^2+1)} = \frac{-x^2-1+9x^2-6x+1}{2(x^2+1)} = \frac{8x^2-6x}{2(x^2+1)} = \frac{4x^2-3x}{x^2+1}$ $\frac{9}{2} - \frac{(x+3)^2}{2(x^2+1)} = \frac{9x^2+9-x^2-6x-9}{2(x^2+1)} = \frac{8x^2-6x}{2(x^2+1)} = \frac{4x^2-3x}{x^2+1}$ $\frac{(3x-1)^2}{2(x^2+1)} \geq 0 \Rightarrow y \geq -\frac{1}{2}$ $\frac{(x+3)^2}{2(x^2+1)} \geq 0 \Rightarrow y \leq \frac{9}{2} \Rightarrow -\frac{1}{2} \leq y \leq \frac{9}{2} \text{ (Not hence gets B1 only.)}$ Turning points occur at equality, so $\left(-3, \frac{9}{2}\right)$ and $\left(\frac{1}{3}, -\frac{1}{2}\right)$ Asymptote is $y = 4$. Intersection with axes at $(0, 0)$ and $\left(\frac{3}{4}, 0\right)$, Intersection with asymptote at $\left(-\frac{4}{3}, 4\right)$ Correct graph.	M1A1 A1 (3) B1 B1 (2) B1B1 (2) B1 (1) B1B1 B1 (3) Total 11

Qn & Part	Solution	Marks
8	$y' = 0 \Rightarrow (x+1)(4x+k) - (2x^2+kx) \times 1 = 0$ $\Rightarrow 4x^2 + (4+k)x + k - 2x^2 - kx = 0 \Rightarrow 2x^2 + 4x + k = 0$ $B^2 - 4AC < 0 \Rightarrow$ no stationary points $\Rightarrow 16 - 8k < 0$ $\Rightarrow k > 2$ for no stationary points. When $k = 4$: Vertical asymptote: $x = -1$ Oblique asymptote: $y = 2x + 2 - \frac{2}{x+1} \Rightarrow y = 2x + 2$ Axes and asymptotes Each branch.	M1 A1 M1A1 A1 [5] B1 M1A1 B1 B1B1 [6] Total 11